

Oncological Emergencies

Recognition and Immediate Management of Life-Threatening Complications

Doc #14 of 15

Domain: Oncology & Cancer
Treatment

Audience: Medical
Professionals

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1. Febrile Neutropenia (FN)

FN is defined as temperature $>38.3^{\circ}\text{C}$ (single) or $>38.0^{\circ}\text{C}$ sustained >1 hour in a patient with $\text{ANC} <0.5 \times 10^9/\text{L}$ (or $<1.0 \times 10^9/\text{L}$ falling). It represents a medical emergency with mortality up to 5-11% (higher in high-risk patients). MASCC score stratifies risk to guide outpatient vs. inpatient management.

Step	Action	Detail
1. Assess	MASCC risk score + clinical assessment	Low risk (MASCC ≥ 21): consider oral ciprofloxacin + amoxicillin-clavulanate as outpatient
2. Cultures	Blood cultures (2 sets: peripheral + CVC if present) + urine MC&S; + CXR	Within 1 hour of presentation
3. Empirical ABx	IV piperacillin-tazobactam 4.5g TDS or cefepime 2g TDS	Meropenem if septic shock or prior resistant organism
4. G-CSF	Continue or start if not already given; do NOT delay antibiotics	Filgrastim or pegfilgrastim — reduces duration of neutropenia
5. Review 48h	De-escalate if cultures negative and patient improving	Add antifungal (echinocandin) if >4 -5 days persistent fever

2. Malignant Spinal Cord Compression (MSCC)

MSCC is a neurological emergency. Delayed treatment is the primary determinant of poor outcome. Neurological function at time of treatment initiation is the strongest predictor of post-treatment ambulatory status.

- Presentation: Back pain (90%) ± radiculopathy, weakness, sensory level, bowel/bladder dysfunction
- Imaging: Urgent whole-spine MRI (within 4 hours of clinical suspicion); plain X-ray insufficient
- Immediate: Dexamethasone 16 mg IV STAT then 4mg QDS; analgesia
- Definitive: Surgery (direct decompressive surgery ± stabilisation) if good PS, single level, solid tumour, paraplegia <48h (Patchell trial — superior to RT alone); RT for multiple levels, haematological malignancy, poor PS
- RT: 8 Gy single fraction equivalent to 30 Gy/10 fr for pain (SCORAD); 20 Gy/5 fr common compromise

3. Tumour Lysis Syndrome (TLS)

Parameter	Laboratory TLS (Cairo-Bishop)	Clinical TLS
Definition	≥2 of: Uric acid ≥476 µmol/L; K ≥6 mmol/L; Phosphate ≥2.1 mmol/L; Ca ≤1.75 mmol/L (within 3 days pre- to 7 days post-treatment)	Lab TLS + ≥1: Cr ≥1.5×ULN; Cardiac arrhythmia; Seizure
High-risk cancers	Burkitt lymphoma, ALL, AML (WBC>100), bulky DLBCL	LDH >2×ULN, bulky disease, renal impairment at baseline
Prevention	IV hydration (3 L/m ² /day); allopurinol 300-900 mg/day (start 2 days before)	Rasburicase 0.2 mg/kg IV for high-risk or established TLS
Monitoring	Every 6-12h: U&E, uric acid, phosphate, Ca, LFTs, urine output	Target urine output ≥100 mL/hr; avoid urinary alkalinisation with rasburicase

4. Superior Vena Cava (SVC) Syndrome

- Causes: NSCLC (most common), SCLC, lymphoma, mediastinal germ cell tumour, thrombosis (CVC-related)
- Features: Facial/arm oedema, plethora, dyspnoea, headache (worse bending forward), dilated collateral veins
- Grading: CTCAE-based; Grade 4 (haemodynamic compromise) = emergency; most are Grade 1-3
- Management: Endovascular SVC stenting (fastest symptomatic relief — hours); dexamethasone 8mg BD; systemic therapy (chemo ± RT) for underlying malignancy
- Biopsy first (if possible and safe) to guide tumour-specific treatment — don't miss SCLC/lymphoma which respond rapidly to chemo

■ TIMING IS CRITICAL: All four oncological emergencies in this document (FN, M SCC, TLS, SVC with haemodynamic compromise) require immediate assessment and treatment initiation within hours. Delay worsens outcomes and increases mortality.